

Akash Chavan

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Experienced **Software Engineer** with over **7 years** of industry experience specializing in **Python-based web applications, AI integration** and **cloud infrastructure**. Proven track record of developing scalable platforms, real-time monitoring systems, and AI-powered applications using **FastAPI, React** and **AWS**. Proficient in database optimization, **RESTful API** development, and containerized deployments. Skilled in implementing Generative AI models for practical applications including **LLMs** and **RAG systems**. **Actively seeking opportunities** to lead complex development projects and build advanced AI-powered solutions.

AWARDS, RECOGNIZATION, CERTIFICATIONS

- Special Recognition in Graduate Studies
Associated with Department of Computer Science, California State University, Los Angeles
- Deep Learning Certification
Associated with Udacity
- National Cyber League (NCL) Spring 2024 Individual Game
Associated with National Cyber League (Ranked 743 out of 7,412 participants)

PROFESSIONAL EXPERIENCE

POLARIS WIRELESS

Software Engineer

Santa Clara, CA

July 2025 - Present

Developed a geolocation analytics platform for law enforcement and security applications, processing real-time location data across simultaneous surveillance operations.

- Designed and built a FastAPI-based REST API platform supporting surveillance features with async request handling and Pydantic-validated request/response schema.
- Architected a route system across feature modules with role-based access control and JWT authentication backed by Redis session store.
- Built real-time target tracking with progressive reveal driven by server-side session, simulating live GPS feed behavior.
- Implemented a multi-case cross-referencing data layer on PostgreSQL, and indexed geospatial queries via PostGIS.
- Designed a typed schema registry with Pydantic validators enforcing field constraints.
- Built an AI-powered data generation agent using LangChain/LangGraph with a 7-tool agentic pipeline (document parsing, schema analysis, code generation, execution, cross-reference validation, deployment, endpoint verification) that produced production-quality test datasets from natural language scenario documents.
- Implemented a multi-LLM provider abstraction layer supporting Anthropic Claude, OpenAI GPT-4o, and Ollama via a unified interface, enabling zero-code provider switching through configuration.
- Designed the agent to generate deterministic Python programs rather than raw data, guaranteeing reproducibility and enabling human-in-the-loop interaction on generated datasets.
- Built a cross-reference validation engine enforcing integrity checks, chronological ordering and cross-case shared entity verification.
- Implemented a retry loop where the agent reads validation failures and self-corrects generated code, achieving first-pass success with a 3-retry calling.
- Built a React + Vite SPA with leaflet map previews, syntax-highlighted code views, real-time SSE progress streaming, and drag-and-drop document upload for the agent workflow.
- Containerized the platform with Docker and orchestrated multi-service deployment (FastAPI backend, React frontend, PostgreSQL, Redis, LangGraph agent service) with health checks and graceful shutdown handling.
- Built an ETL pipeline to ingest and process large-scale telecommunications data using Pandas and NumPy for transformation, DuckDB for analytical queries across millions of rows and Python multiprocessing to parallelize CPU-bound tasks.

BETTER ANGELS**Software Engineer****Los Angeles, CA****Sept 2024 - July 2025***Built AI-powered homeless service platform used by LA caseworkers to manage clients, match them to shelters and coordinate care.*

- Built an AI service navigator using LLM function calling over WebSocket chat that lets caseworkers find shelters, legal aid, food banks, and other services through natural language instead of manual filters.
- Implemented hybrid search combining pgvector semantic similarity, PostGIS geospatial filtering, and PostgreSQL array attribute matching, merged using Reciprocal Rank Fusion.
- Set up a Celery-based weekly ingestion pipeline from Data platform - only generates embeddings when provider descriptions actually change, keeping costs under.
- Designed a 3-layer permissions engine using SQL CTEs with SQLAlchemy async, field-level filtering on every response.
- Built event-sourced notes with an append-only event log in PostgreSQL and two-phase time-travel revert, served via FastAPI with React/TypeScript frontend.
- Created an AI-assisted client intake flow using Tesseract OCR + LLM for document extraction, feeding into a declarative rule engine for service qualification.
- Implemented shelter bed reservations with concurrency and a Celery reconciliation task every 15 minutes to prevent overbooking.
- Wrote integration tests running against a real PostgreSQL instance with PostGIS and pgvector extensions, plus concurrent reservations stress tests.
- Containerized application with Docker Compose for local development and deployed to AWS ECS for production, with separate service definitions for FastAPI server, Celery workers, Celery Beat scheduler and imgproxy
- Used S3 for document uploads and shelter photos, with imgproxy handling image resize and thumbnail generation on the fly.
- Set up separate Redis instances for Celery task queue and application-level caching, with session storage for WebSocket chat conversation and 24-hours session expiration.

FOSSEE, IIT BOMBAY**Software Engineer****Mumbai, India****Feb 2019 - May 2022***Built and maintained scalable e-learning platform serving educational content to thousands of users with advanced code evaluation capabilities*

- Implemented a scalable e-learning platform handling 10,000+ concurrent users using Django, combining video lessons, automated code evaluation, and interactive quizzes with Python, C, C++, Java, R, Scilab and Bash Support.
- Designed a secure, distributed code execution system using Django Channels and Tornado web Server to safely evaluate student-submitted code in real-time with process isolation and resource constraints.
- Implemented a leader-follower architecture for code evaluation using process pools to handle concurrent requests, implementing job queues for load balancing and efficient resource utilization, reducing response time by 50%.
- Optimized Django ORM queries using select_related and prefetch_related, resolving N+1 query issues and reducing page load times by 60%.
- Implemented asynchronous background task processing using Django Celery and Redis for notifications, scheduled reminders and resource-intensive operations, improving system reliability.
- Migrated frontend from Django templates to [Vue.js](https://vuejs.org/), creating a responsive single-page application with improved user experience and real-time updates, increasing user engagement by 40%.
- Created REST APIs using Django REST Framework to support mobile clients and third-party integrations, with proper authentication and rate limiting.
- Implemented analytics dashboard for instructors to monitor student progress, identify learning gaps and optimize course content based on performance data.
- Implemented timetabling solutions using Pandas/PySpark, NumPy and Bash, reducing scheduling errors and enhanced operational efficiency
- Implemented automated configuration management using Docker containers and orchestration tools for distributed system deployment and maintenance.

Developed remote access solution for Single Board Heater System enabling distributed laboratory experiments with real-time monitoring and control capabilities

- Architected a distributed remote lab platform connecting physical SBHS boards via 5 Raspberry Pi agents to a central Django server, enabling students to remotely perform control system experiments over the internet.
- Designed a multi-layer experiment coordination system where the Django server relays heat/fan/temperature commands through RPi agents to physical boards over USB serial, with JWT auth, slot-based access control and experiment logging.
- Designed a multi-protocol client bridge enabling Scilab integration via direct HTTP, allowing students to control real hardware from Scilab scripts with real-time plotting.
- Deployed a production system on a lab server with Caddy reverse proxy, PostgreSQL, Django and React behind Firewall with Raspberry Pis running agents natively over USB serial.
- Built an observability stack with structured JSON logging, correlation IDs tracing requests across server, Prometheus metrics and Grafana dashboards.
- Wrote comprehensive test suites across all servers - unit tests with mocked serial communication, integration tests, and end-to-end API tests ensuring the full chain works correctly.
- Implemented database-level concurrency safety for slot booking using row-level locking to prevent double-booking race conditions in a system.

PROJECTS

AI-powered document learning platform

- Built an AI-powered document analysis platform using [Next.js](#), React 19, TypeScript and PostgreSQL.
- Implemented a microservices architecture with separate FastAPI service for document processing and embedding generation.
- Designed RESTful APIs handling user authentication, document management, realtime chat and vector similarity search.
- Integrated OpenAI GPT-4 for intelligent document Q&A with context-aware responses and proper citation formatting.
- Implemented Retrieval Augmented Generation (RAG) pipeline using sentence-transformers for semantic document search.
- Built custom embeddings service using FastAPI and sentence-transformers for PDF document vectorization.
- Designed and implemented pgvector extension for PostgreSQL to enable efficient semantic similarity search.
- Deployed scalable application infrastructure using Docker containers and AWS services (S3, EC2)
- Implemented secure file storage and retrieval system using AWS S3 with presigned URLs.
- Configured CI/CD pipeline and container orchestration for development and production environments.
- Designed system to handle concurrent user uploads and document processing through asynchronous job queues.
- Implemented efficient document chunking and processing pipeline handling PDFs up to 10MB.

Email Digital Twin - AI-powered Email Response System

- Developed a digital twin application using FastAPI backend and [Next.js](#) frontend that analyzes incoming emails with PDF attachments and generates responses mimicking user writing style.
- Implemented OAuth 2.0 Gmail integration with secure session management, enabling seamless email retrieval, thread analysis, thread analysis and PDF attachment processing using PyPDF2 for text extraction.
- Created AI-powered analysis pipeline using OpenAI GPT-4o-mini to process email context, extract key insights and generate contextually relevant responses based on conversation history and attachment content.
- Build scalable microservices architecture with modular service layers (Gmail API, PDF processing, AI analysis) and RESTful API endpoints supporting both individual email and thread-based operations.

SKILLS

- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB
- **Programming Languages:** Python, JavaScript/TypeScript, Java, SQL
- **Backend Frameworks:** Django, FastAPI, [Node.js](#), [Express.js](#), Flask, GraphQL
- **Frontend Technologies:** React, Vue, [Next.js](#), TailwindCSS, HTML, CSS
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, CloudFront), Docker, Kubernetes, CI/CD, GitHub Actions
- **AI/ML:** LangChain, LangGraph, Vector Databases, RAG Systems
- **Testing:** Test-driven development (TDD), PyTest, UnitTest, Jest, Mocha, Chai
- **AI Tools:** Claude Code, Cursor, Codex, Cline, Copilot, Kilo Code

EDUCATION

CALIFORNIA STATE UNIVERSITY, LOS ANGELES

MS in Computer Science (GPA: 4/4)

Los Angeles, US

May 2024

DR. BABASAHEB AMBEDKAR MARATHWADA UNIVERSITY

B.E Computer Science and Engineering

Aurangabad, India

May 2016